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June 16, 2026

Prof. Djamila Aouada
Computer Vision, Imaging, and Machine Intelligence (CVI²)
Interdisciplinary Centre for Security, Reliability and Trust (SnT)
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Dear Prof. Aouada,

I am writing to express my strong interest in the Doctoral Researcher position focusing on Vision-Language-Action (VLA) Models in Autonomous Driving within the CVI² research group at SnT. As an AI & Data Science Engineer with a comprehensive background in multi-modal deep learning architectures, advanced computer vision, and high-performance inference, I am highly motivated to contribute to your industrial collaboration with POST Luxembourg to enhance the robustness and long-horizon reasoning of autonomous systems.

In my recent work on the "Cardiovascular Medical Digital Twin," I engineered **Medical_MambaTS**, a state space model architecture designed for sub-millisecond, linear-time inference on the edge. This required tackling the exact challenges faced in autonomous driving: real-time, low-latency processing of multi-dimensional data streams (100Hz VitalDB) while maintaining strict predictive safety margins. Furthermore, my strong background in PyTorch and C++ allowed me to build a zero-copy PyArrow pipeline that prevents GPU starvation—a crucial architectural necessity for training large-scale VLA models on extensive autonomous driving datasets.

I also have direct experience with multi-modal reasoning and handling "rare and novel scenarios." By designing a Missingness-Aware Gate and a physics-informed DeepONet (PI-DeepONet) utilizing PyTorch Autograd, my models learned to dynamically adapt to sudden sensor failures and unseen physiological states without retraining. I am eager to transition these advanced architectural paradigms—combining semantic language embeddings (SBERT) and robust spatio-temporal reasoning—to the domain of autonomous driving perception and planning.

Enclosed are my CV, a description of my previous projects, and my GitHub portfolio highlighting my computer vision and deep learning implementations. I would be honored to join your dynamic, international team at the University of Luxembourg and contribute to securing and advancing VLA-based driving systems.

Sincerely,

Ismail Aissaoui